

MATH 215 SI

Session 3

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Lectures 9 - 11

- ▶ 14.1 Functions of Several Variables
- ▶ 14.3 Partial Derivatives
- ▶ 14.4 Tangent Planes and Linear Approximations

14.1 Functions of Several Variables

A function f of two variables is a rule that assigns to each ordered pair (x, y) in a set D a unique real number $f(x, y)$

- ▶ The set D is the domain of f , and the set of values f takes on is the range
- ▶ The graph of f is the surface $\{(x, y, z) \in \mathbb{R}^3 \mid z = f(x, y), (x, y) \in D\}$

14.1 Functions of Several Variables

cont.

Level Curves and Level Surfaces

- ▶ The level curves of $f(x, y)$ are the curves $f(x, y) = k$ for constants k in the range of f
- ▶ A contour map is a collection of level curves
- ▶ The level surfaces of $f(x, y, z)$ are the surfaces $f(x, y, z) = k$

Exercises

14.1 Functions of Several Variables

- ▶ Sketch a few level curves of $f(x, y) = y - x^2$
- ▶ Describe the level surfaces of $g(x, y, z) = x^2 + y^2 - z^2$ for $k > 0$, $k = 0$, and $k < 0$

14.3 Partial Derivatives

The partial derivative of f with respect to x at (a, b) is given by

$$f_x(a, b) = \lim_{h \rightarrow 0} \frac{f(a+h, b) - f(a, b)}{h}$$

- ▶ To find f_x , treat y as a constant and differentiate with respect to x
- ▶ To find f_y , treat x as a constant and differentiate with respect to y
- ▶ Notation: $f_x = \frac{\partial f}{\partial x} = \frac{\partial z}{\partial x} = D_x f$
- ▶ $f_x(a, b)$ is the slope of the tangent line to the trace of the graph of f in the plane $y = b$

14.3 Partial Derivatives

cont.

Higher Derivatives

- ▶ Second partial derivatives: $f_{xx}, f_{xy}, f_{yx}, f_{yy}$
- ▶ Note that $f_{xy} = (f_x)_y = \frac{\partial^2 f}{\partial y \partial x}$
- ▶ Clairaut's Theorem: if f_{xy} and f_{yx} are both continuous on a disk containing (a, b) , then $f_{xy}(a, b) = f_{yx}(a, b)$

Implicit Differentiation

- ▶ If z is defined implicitly by an equation in x, y, z , differentiate both sides with respect to x (treating y as constant and z as a function of x and y), then solve for $\frac{\partial z}{\partial x}$

Exercises

14.3 Partial Derivatives

Find or estimate $f_x(0,0)$ and $f_y(0,0)$ for each of the following functions

- ▶ $f(x,y) = y^2 \cos(1+x-y^2x)$
- ▶ g described by the table below

$x \backslash y$	-2	-1	0	1	2
-2	6	9	9	9	10
-1	12	16	18	19	20
0	20	22	25	27	30
1	28	36	43	47	48
2	35	49	55	61	66

Exercises

14.3 Partial Derivatives cont.

Suppose $f(x, y)$ is differentiable with continuous derivatives, and

$$f(2, 5) = 7 \quad f_x(2, 5) = 3 \quad f_y(2, 5) = -2$$

- ▶ Let C be the curve of intersection of the plane $x = 2$ and the surface $z = f(x, y)$
- ▶ Find a parametric equation of the line that lies in the plane $x = 2$ and is tangent to C at $(2, 5, 7)$

14.4 Tangent Planes and Linear Approximations

Suppose f has continuous partial derivatives. The tangent plane to the surface $z = f(x, y)$ at the point $(a, b, f(a, b))$ is given by

$$z - f(a, b) = f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

- ▶ Compare this to the tangent line $y - f(a) = f'(a)(x - a)$ from single variable calculus
- ▶ The linearization of f at (a, b) is
$$L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$
- ▶ For (x, y) near (a, b) we have the linear approximation $f(x, y) \approx L(x, y)$

14.4 Tangent Planes and Linear Approximations

cont.

Differentiability and Differentials

- ▶ If f_x and f_y exist near (a, b) and are continuous at (a, b) , then f is differentiable at (a, b)
- ▶ Existence of the partial derivatives alone does **not** guarantee differentiability
- ▶ The total differential of $z = f(x, y)$ is given by $dz = f_x(x, y) dx + f_y(x, y) dy$
- ▶ dz approximates the actual change Δz when $dx = \Delta x$ and $dy = \Delta y$ are small

Exercises

14.4 Tangent Planes and Linear Approximations

Suppose $g(x, y) = x + \ln(5x^2 - 4y^2)$

- ▶ Find an equation for the tangent plane to the surface $z = g(x, y)$ at the point $(1, 1, 1)$
- ▶ Find the linearization of g at $(1, 1)$, and use it to approximate $g(1.02, 0.99)$

Bonus

Which of the following equations does $z = f(x + t) + g(x - t)$ satisfy for all differentiable single-variable functions $f(s)$ and $g(s)$?

► (a) $\frac{\partial z}{\partial x} + \frac{\partial z}{\partial t} = 0$

► (b) $\frac{\partial z}{\partial x} = \frac{\partial z}{\partial t}$

► (c) $\frac{\partial z}{\partial x} + \frac{\partial^2 z}{\partial t^2} = 0$

► (d) $\frac{\partial z}{\partial x} = \frac{\partial^2 z}{\partial t^2}$

► (e) $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial t^2} = 0$

► (f) $\frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 z}{\partial t^2}$

Fin

This concludes session 3.